

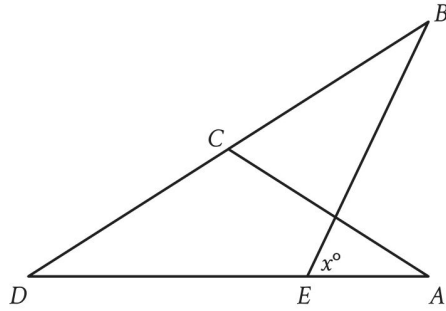
SAT Math

Lines, Angles, and Triangles 3

Question # ID
3.1 6d99b141

Answer

83

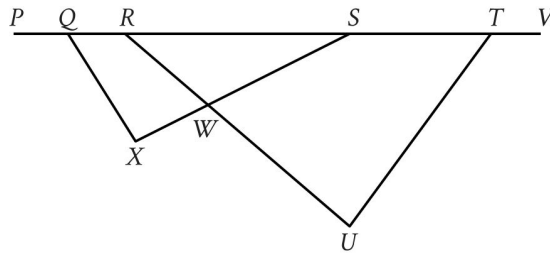


Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is 45° , and the measure of angle ACD is 104° . What is the value of x ?

3.2 e10d8313

123



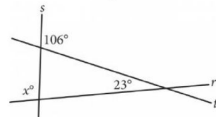
Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

3.3 f88f27e5

97

Intersecting lines r , s , and t are shown below.



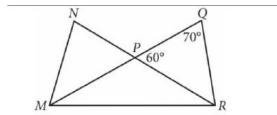
What is the value of x ?

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Question # ID
3.4 947a3cde

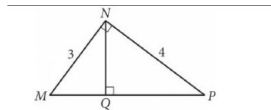
Answer
30



In the figure above, $\overline{MN}$ and $\overline{QR}$ intersect at point P , $NP = QP$, and $MP = RP$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

3.5 740bf79f

C

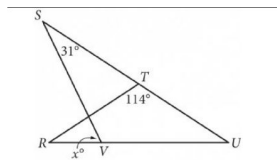


In the figure above, what is the length of NQ ?

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

3.6 bd7f6e30

C

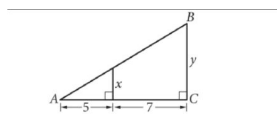


In the figure above, $RT = TU$.
What is the value of x ?

- A. 72
- B. 66
- C. 64
- D. 58

3.7 eeb4143c

10/3,
15/4,
25/6



Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

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Question # ID
3.8 5b4757df

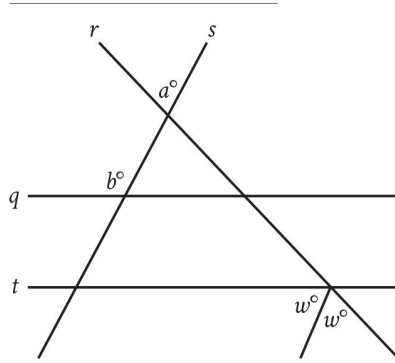
Answer

44/3

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

3.9 17912810

50.5

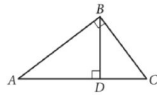


Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

3.10 6a3fbec3

4.5



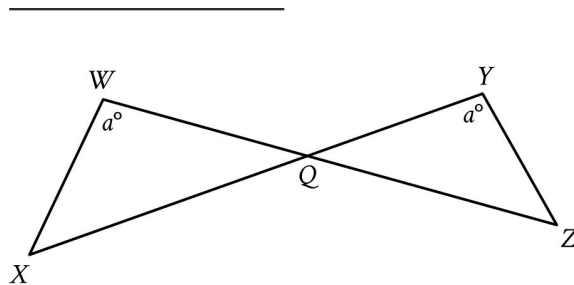
Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$.

What is the length of $\overline{DC}$?

3.11 345cc36a

54



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

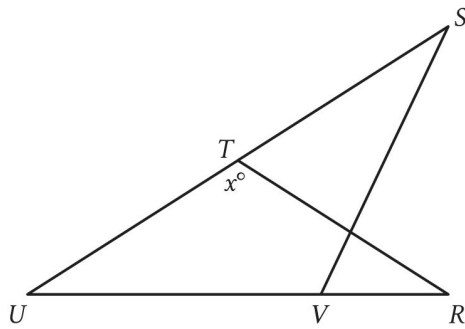
SAT Math

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Question # ID
3.12 2d2cb85e

Answer

156



Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

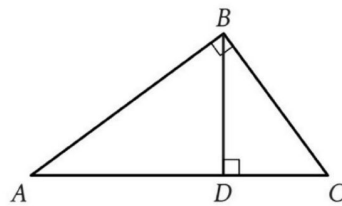
3.13 f7dbde16

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

D

3.14 abcd0011



Note: Figure not drawn to scale.

In the figure shown, $AD = \frac{121}{3}$ and $AB = \frac{11\sqrt{130}}{3}$. What is the length of $\overline{DC}$?

3

SAT Math

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Question # ID

Answer

3.15 322a6dfe

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

D

- A. 5°
- B. 27°
- C. 45°
- D. 135°

3.16 fecc446d

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

A

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360